Yaowen Shi

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Education

South China University of Technology (SCUT)

B.S. in Big Data Management and Application *Sep. 2023 – Jun. 2027 (Expected)*

- Academic standing: GPA **3.94/4.00**; average score: **92.12**; major rank: **3/156**
- Selected courses: Calculus I/II (**99, 100**); Linear Algebra and Analytic Geometry (**100**); Operations Research (**99**); Python Data Analysis (**97**); Probability and Statistics (**96**); Large Language Models and GenAI (**96**); Data Structures (**95**); **ranked 1st in 11 courses**.

Research Interests

I work on LLM agents and agentic reinforcement learning. My research focuses on large-scale synthesis of verifiable agentic training data—building executable environments that yield tasks with automatic correctness checks—and on using execution feedback as reward to train agents that plan, act, and self-correct over long horizons. I am broadly interested in the co-design of environments, data synthesis, and RL algorithms for general-purpose agents.

Publications

3D Code Factory: Scaling Verifiable 3D Tasks for Coding Agents

Yaowen Shi, *Jinxiu Liu*, *Weiyang Liu*, *Yandong Wen*

Under Review (ICLR 2027).

- Introduced a fact-anchored synthesis framework that co-generates implementation-agnostic 3D coding tasks and multimodal rubrics from shared scene facts, verifying spatial and mechanical properties from runtime scene state and perceptual properties from rendered views instead of fixed reference matching.
- Built 3D Live Bench—36 core tasks in Three.js and Blender executed by command-line coding agents (Codex, Claude Code, OpenCode) with MCP 3D tools—and scaled the pipeline to **488** verified tasks over 500 scenarios through a nine-dimensional quality review.
- Evaluated 17 frontier model–scaffold configurations, showing that near-perfect program execution hides rubric compliance of only 0.40–0.80 and that scaffold choice shifts task success and token cost as much as the base model does; fine-tuning Qwen3-8B on 786 synthesized multi-turn trajectories raised Blender execution from **20.0%** to **60.0%**.

SymGround: Visual Program Synthesis with Symbol-to-Vision Grounding

Yaowen Shi, *Jinxiu Liu*

Under Review (ARR).

- Formulated visual-to-program credit assignment—grounding rendered residuals in editable program symbols—as a core bottleneck in executable visual synthesis.
- Built a relation-grounded revision loop with visual attribution, evidence-guided editing, and edit-render memory, representing each edit as a testable symbol-to-vision hypothesis.
- Evaluated on SlideBench, BlenderGym, and BlenderBench; outperformed VIGA-class baselines, raising the BlenderBench aggregate score from **89.12** to **137.99** and cutting invalid executions from **7.47%** to **0.16%** with fewer tokens per task.

Experience

Visiting Student, Westlake University May 2026 – Present

Advised by Yandong Wen, Assistant Professor at Westlake University

- Building a visual "code factory" that automatically constructs executable environments and verifiable tasks for 3D (Blender Python) and web (Three.js) code generation, extending SWE-bench-, SWE-bench Pro-, and GDPval-style task construction with minimal human annotation.
- Developing reinforcement learning for visual code agents on top of these environments, using execution- and render-based verification as scalable reward signals.
- Built a benchmark for procedural 3D code generation in Three.js and Blender, evaluating how agents plan, write, debug, and refine code to produce visual artifacts such as 3D scenes and web pages.

Multimodal Agents for Visual Creation Mar. 2025 – Present

Collaborator: Jinxiu Liu, Research Scientist at Nex-AGI

- Developed SymGround for executable slide and Blender-scene synthesis, separating visual attribution from localized code editing and using edit-render evidence to avoid unsupported retries and broad rewrites.
- Built multimodal generation-agent workflows with NexAU and ComfyUI, defining tool-using agents, skills, and workflow primitives for image/video generation and editing; achieved state-of-the-art results on ComfyBench.
- Explored controllable generation methods including self-reflection/RL for flow-matching text-to-image models, AMF-based video motion transfer with DiT models, and VQ-VAE/adjacency-aware tokenization for 3D meshes.

Projects

OpenDinq | *Evidence-backed AI-native profiles, card workspaces, and explainable people discovery* GitHub

- Built an open-source product alpha that turns public information about a person into reviewable profiles with evidence-backed claims, curated cards, and public profile pages.
- Implemented ingestion from GitHub, websites, OpenAlex, arXiv, ORCID, links, and notes; normalized, deduplicated, and quality-scored claims for explainable people discovery.

Awards

Meritorious Winner (Top 7%), Mathematical Contest in Modeling (MCM) / Interdisciplinary Contest in Modeling (ICM) 2025

Second Prize (Top 15%), Contemporary Undergraduate Mathematical Contest in Modeling (CUMCM), Guangdong Division 2024

Technical Skills

Languages: C++, Python, SQL, JavaScript, HTML, CSS
ML: PyTorch, Transformers, diffusers
Developer Tools: Git, Docker
English: CET-6